Facial Recognition Using OpenFace Inception Network

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I. Introduction

The phrase "face recognition" describes tasks that behave very differently in practice. *Verification* is one-to-one: a subject claims an identity and the system confirms or denies it against one stored template. *Closed-set identification* is one-to-N: the system decides which of N enrolled people a photograph depicts, knowing the answer is among them. *Open-set identification* additionally permits the answer "nobody enrolled," and is the case that matters for access control [7].

Published benchmark figures overwhelmingly report the first. OpenFace's $92.92\% \pm 1.34\%$ for the model used here, and the FaceNet paper's 99.63%, are both measured on the standard LFW list of 6,000 pairs, which is a verification protocol; neither reports an identification metric. The same is true of DeepFace [2] and of the LFW leaderboard generally, which is why MegaFace [6] was later constructed specifically to measure identification at scale.

We build a pipeline on the pretrained OpenFace `nn4.small2` network [3] and evaluate all three tasks on Labeled Faces in the Wild (LFW) [4] using identical embeddings. On the same embedding we measure 89.2% verification, 58.7% closed-set rank-1 on a 150-identity gallery, and 14.7% open-set detection and identification rate at 1% false accept rate. Our contributions are:

1) A like-for-like measurement of all three tasks on one embedding, with preprocessing validated on a tuning split disjoint from the split used to report accuracy.
2) An order-statistics account of the closed-set gap, checked for consistency across gallery sizes from 2 to 300. A per-probe formulation tracks measurement to within 2.8 points up to $N = 150$; the pooled formulation, differing only in where the average is taken, errs by a factor of 55. The informative result is the sign and growth of its residual, which isolates where the independence assumption fails.
3) An ablation separating two preprocessing effects: on the tuning split, shrinking the crop window gains 17.1 points, and displacing a window downward by 18 pixels gains 12.7 to 14.6 points at fixed window size.
4) A measured improvement predicted by the model: multi-shot enrolment raises rank-1 from 52.0% to 69.3%, and we verify the mechanism is the one the model implies rather than a general contraction of the embedding space.

Reproducibility. All reported numbers and figures are produced by `paper_experiments.py` in the accompanying repository,[1] which is seeded throughout and prints each result under the table it feeds. The one exception is the detector comparison of Section VI-B, which requires an external model file not redistributed with the repository.

II. Background

A. Metric learning

A softmax classifier with one output per identity must be retrained whenever someone is enrolled, its output layer grows with the roster, and it cannot express "none of the above," since its outputs are normalised over known classes only.

Deep metric learning instead learns an embedding $f : \mathcal{I} \to \mathbb{R}^{128}$ in which Euclidean distance encodes identity. Enrolment costs one forward pass, and open-set rejection becomes a threshold test.

B. Triplet loss

FaceNet [1] trains such an embedding on triplets of an anchor A, a positive P (same person) and a negative N^-, requiring $\|f(A) - f(P)\|_2^2 + \alpha \le \|f(A) - f(N^-)\|_2^2$, which hinged and summed over a batch gives

$$\mathcal{L} = \sum_i \max\big(\|f(A_i) - f(P_i)\|_2^2 - \|f(A_i) - f(N_i^-)\|_2^2 + \alpha,\ 0\big). \tag{1}$$

The margin α is essential: without it $f(x) = c$ satisfies every constraint at zero loss. Since the number of candidate triplets grows cubically in the training set size, training mines *semi-hard* negatives online, satisfying $\|f(A) - f(P)\|_2^2 < \|f(A) - f(N^-)\|_2^2 < \|f(A) - f(P)\|_2^2 + \alpha$: these violate the margin, and so produce gradient, without being the pathological hardest cases that destabilise training. Later work replaces sampled triplets with angular margins imposed directly on the hypersphere, as in CosFace [9] and ArcFace [10].

C. Embedding geometry and distance convention

L^2 normalisation places embeddings on the unit hypersphere, so distances are bounded to $[0, 2]$ rather than $[0, 1]$, and $\|f(x_i) - f(x_j)\|_2^2 = 2 - 2\cos\theta_{ij}$ makes squared Euclidean distance and cosine similarity interchangeable for ranking.

[1] https://github.com/fistfulofyen/Face_sys_LFW

All distances and thresholds here are unsquared L^2. This matters because OpenFace's own default threshold of 0.99 is on *squared* L^2, equivalent to 0.995 unsquared, numerically close to the operating point we obtain. The proximity is a coincidence of this dataset, not a rediscovery of their default; under the squared convention our threshold would be 0.984.

III. Method

A. Network

The network is an Inception-style CNN [5] with 3,733,968 trainable parameters over 155 layers, comprising 37 convolutions, 37 batch-normalisation layers and 7 depth concatenations. Keras reports a total of 3,743,280; the difference of 9,312 is exactly the non-trainable batch-normalisation moving means and variances, two per each of 4,656 normalised channels. The trainable count matches OpenFace's published figure, a useful check that the port is faithful.

It maps a channels-first $(3, 96, 96)$ RGB tensor to a 128-dimensional unit-norm vector. A three-block stem downsamples to 12×12, seven Inception stages reduce this to 3×3 while growing depth to 736 channels, and a dense layer with L^2 normalisation produces the embedding. The 1×1 branches inside each stage act as bottlenecks that hold the parameter count at 3.7M.

No training is performed. The network is a frozen feature extractor with weights ported from OpenFace, trained on CASIA-WebFace and FaceScrub, roughly 500,000 images. This is about $400\times$ less data than the private corpus behind FaceNet, so the appropriate baseline for this reproduction is OpenFace's $92.92\% \pm 1.34\%$, not FaceNet's 99.63%.

B. Weight transfer

Parameters ship as 226 flat CSV exports of the original Torch model, of which 224 are used: two tensors for each of 37 convolutions, four for each of 37 batch-normalisation layers, and two for the dense layer. The two unused files are the moving mean and variance exported for `inception_5a_3x3_conv2`, a convolution that has no batch-normalisation layer in this architecture. Restoring them requires transposing convolution kernels from Torch's $(\text{out}, \text{in}, h, w)$ layout to TensorFlow's $(h, w, \text{in}, \text{out})$. One inherited quirk had to be preserved: the first stem block uses $\varepsilon = 10^{-3}$, the framework default, because it was originally instantiated without an explicit value, while all 36 others use 10^{-5}. Our implementation reproduces the embeddings of the Keras-OpenFace reference port to 0.000e+00 maximum element-wise difference, with all 75 weight-bearing layers matching.

C. Decision rules

The classifier is non-parametric. Verification accepts if $\|f(x) - e_{\text{claim}}\|_2 < \tau$. Closed-set identification returns $\arg\min_k \|f(x) - e_k\|_2$. Open-set identification returns that argument only if the corresponding minimum falls below τ, and otherwise reports the probe as not enrolled. Following [7], we score open-set performance by the detection and identification rate (DIR), the fraction of enrolled probes both correctly ranked first and accepted, measured at a fixed false accept rate (FAR) on never-enrolled probes.

IV. Experimental Setup

LFW contains 13,233 photographs of 5,749 individuals from online news sources. We use the `lfw-deepfunneled` variant, already aligned by the dataset's congealing procedure, which matters in Section VI-B. Of its identities 70.8% have exactly one photograph; these can populate the mismatched half of the pairs list and serve as open-set impostor probes, but cannot be closed-set probes.

Avoiding selection bias. The crop hyperparameters are worth more than 30 accuracy points, so choosing them on the evaluation data would inflate the result, and cross-validating τ gives no protection against that. We partition the official pair list into disjoint halves and draw a 600-pair *tuning* split and a 600-pair *test* split, sharing no pairs, though 99 individual images appear in both because LFW reuses photographs across pairs.

One caveat is due here. The crop constant was fixed in the codebase during earlier development, before this partition existed, using LFW pairs that are not guaranteed disjoint from the present test split. We therefore do not claim it was chosen under a clean protocol. What Experiment 1 does is *validate* it: it re-runs the whole crop family on the tuning split and confirms the shipped window sits within the noise floor of the best available configuration. The report describes the configuration the released code actually runs, rather than substituting whichever window happened to score highest, and the residual optimism from the constant's origin is acknowledged in Section VI-C.

We use 600 of the 3,000 available pairs per half to keep the nine-configuration ablation tractable on CPU; this is the direct cause of the wide confidence intervals reported throughout, and a full-list evaluation would tighten them.

Estimators. Three quantities appear in the literature as "accuracy." *CV accuracy* cross-validates τ over ten folds. *Fixed-τ accuracy* is the accuracy at one stated threshold. *Best-τ accuracy* is the maximum over a sweep on the same data, and is optimistic. Every accuracy here is CV accuracy unless stated. Note that our folds are random partitions of our own 600-pair sample, not LFW's ten predefined identity-disjoint folds; we follow the spirit of the protocol, not its letter, and Section VI-C returns to the consequence.

Identification draws a pool of 300 multi-photograph identities, enrolled with one photograph and probed with a second. For each gallery size N we average over 200 random sub-galleries, so all points are estimated identically; the sole exception is $N = 300$, where only one gallery exists.

V. Results

A. Preprocessing: placement matters as much as size

Table I separates two effects, both measured on the tuning split so the comparison is internally consistent. Shrinking the

TABLE I
CROP ABLATION, TEN-FOLD CV ACCURACY ON BOTH SPLITS. SELECTION USED THE TUNING COLUMN ONLY. NOTE THAT THE RANKING OF THE THREE SHIFTED WINDOWS *reverses* ACROSS SPLITS, DISCUSSED IN THE TEXT.

Crop window	Tuning split	Test split
None (full 250 px frame)	57.7%	56.8%
160 px, centred	64.5%	66.8%
128 px, centred	66.2%	65.8%
110 px, centred	72.0%	72.2%
96 px, centred	74.8%	74.3%
92 px, centred	73.7%	73.2%
96 px, 18 px lower	87.5%	89.7%
92 px, 18 px lower (shipped)	**88.3%**	**89.2%**
88 px, 18 px lower	88.7%	88.8%

window from the full frame to 96 pixels gains 17.1 points, with diminishing returns; the further step to 92 pixels is slightly negative, which is itself within the noise discussed below.

The second effect is best read at fixed window size, so that only position varies. Displacing the 96 px window 18 pixels down gains 12.7 points, and the 92 px window 14.6 points. The effect is therefore $+12.7$ to $+14.6$ points regardless of which window is used, which is a stronger statement than any single pairing, and it means crop *placement* rivals crop size. A size-only search would have stopped at 74.8%.

The explanation is that a centred window on a deep-funnelled LFW image sits over the forehead and hairline, whereas the weights were fitted on landmark-aligned crops centred nearer the eyes, nose and mouth.

The size choice within the shifted family is noise. The three shifted windows span only 1.2 points on the tuning split and 0.9 on the test split, against a fold-to-fold spread of ± 3.3 points, and their ranking is exactly reversed between the two: worst-to-best is $96 < 92 < 88$ px on tuning and $88 < 92 < 96$ px on test. The window that scores highest on tuning (88 px) is the worst of the three on held-out data. No conclusion about crop size within this family is supportable; only the 18 px downward displacement is a real effect, and it is large enough to be unambiguous.

The pipeline ships the $(92, +18)$ window. Note this is the window whose *centred* variant scored worst of the three sizes; once the offset is applied the ordering no longer holds, which is the same noise argument made above. Evaluated once on the held-out test split it gives $\mathbf{89.2}\% \pm \mathbf{3.3}\%$ CV accuracy ($535/600$ pairs, Wilson 95% CI $[86.4\%, 91.4\%]$), AUC 0.97, EER 10.0%, at $\tau = 0.992$. This is about 4 points below OpenFace's reported $92.92\% \pm 1.34\%$. We do not claim to have closed that gap: our $\pm 3.3\%$ is a fold-to-fold spread over 600 pairs, not a standard error, and OpenFace's interval is over the full 6,000, so the two are not directly comparable. The shortfall is consistent with our smaller evaluation and coarser alignment.

A control confirms the crop effect is framing and not resampling: repeating the uncropped measurement with nearest-neighbour, bilinear, bicubic, area and Lanczos interpolation spans only 56.8% to 59.5%, a 2.7-point range against the ± 3.3-point fold spread, so the choice of resampling filter is not distinguishable from noise.

TABLE II
FIXED-τ OPERATING POINTS ON THE HELD-OUT TEST SPLIT.

Threshold τ	FAR	FRR	Fixed-τ accuracy
0.700	1.0%	51.7%	73.7%
0.850	2.0%	26.7%	85.7%
0.950	7.0%	14.3%	89.3%
0.992 (CV-selected)	**8.3%**	**12.0%**	**89.8%**
1.100	19.7%	4.3%	88.0%
1.250	42.7%	0.7%	78.3%
Best-τ accuracy (optimistic, same data): 90.5%			

TABLE III
ORDER STATISTICS AND THE TWO PREDICTIVE MODELS. OBSERVED VALUES ARE MEANS OVER 200 *randomly resampled* SUB-GALLERIES ($N = 300$ IS A SINGLE GALLERY); "$\pm$" IS THE STANDARD DEVIATION ACROSS SUB-GALLERIES. THESE ARE AVERAGES OVER MANY GALLERIES, NOT THE ACCURACY OF THE ONE GALLERY THE PIPELINE ENROLS, WHICH IS REPORTED SEPARATELY IN THE TEXT.

N	$\mathbb{E}[\min d_{\text{imp}}]$	Observed	Eq. (2)	Eq. (3)
2	1.28	$98.3 \pm 9.2\%$	96.9%	96.9%
5	1.07	$91.3 \pm 13.4\%$	90.9%	88.3%
10	0.99	$85.0 \pm 10.7\%$	85.0%	75.5%
25	0.88	$75.2 \pm 8.7\%$	75.8%	47.3%
50	0.82	$67.2 \pm 6.1\%$	68.5%	21.7%
100	0.77	$59.1 \pm 4.4\%$	61.3%	4.6%
150	0.75	$54.7 \pm 3.1\%$	57.5%	1.0%
300	0.70	46.7%	51.9%	0.01%

B. Verification operating points

Table II corrects a common assumption. A threshold of 0.7, frequently inherited from tutorial implementations, is highly conservative here: it admits 1.0% of impostors while rejecting 51.7% of genuine pairs. The cross-validated optimum sits near the equal error rate. Which point to prefer is a policy question, and the choice moves accuracy by more than 15 points. The full operating characteristic is shown in Fig. 2.

C. Closed-set identification and an order-statistics model

Verification compares against one template; identification competes against every enrolled person, so the deciding quantity is the *minimum* over $N - 1$ impostor distances. The mean impostor distance is 1.28 regardless of N, yet the expected nearest impostor falls from 1.28 at $N = 2$ to 0.70 at $N = 300$ (Table III). That descending bar is what erodes accuracy.

Let p_i be the probability that a randomly chosen wrong identity is nearer to probe i than its own gallery entry. Under independence across distractors,

$$\hat{r}_{\text{probe}}(N) = \frac{1}{n} \sum_{i=1}^{n} (1 - p_i)^{N-1}, \tag{2}$$

$$\hat{r}_{\text{pool}}(N) = (1 - \bar{p})^{N-1}, \tag{3}$$

where $\bar{p}$ is the mean of p_i. The two differ only in whether the average precedes or follows exponentiation.

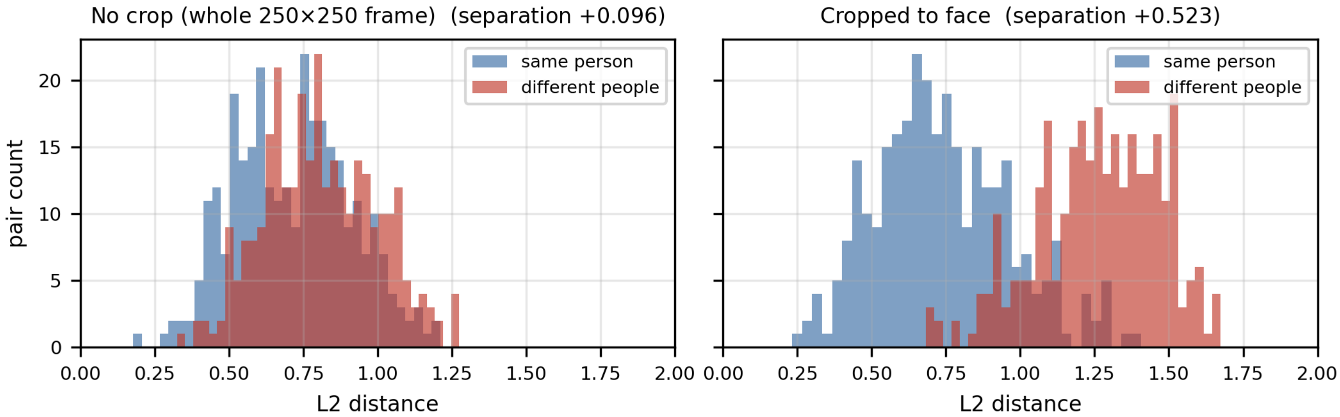

Fig. 1. Matched and mismatched distance distributions on the held-out test split. Cropping separates the distributions by roughly a factor of five.

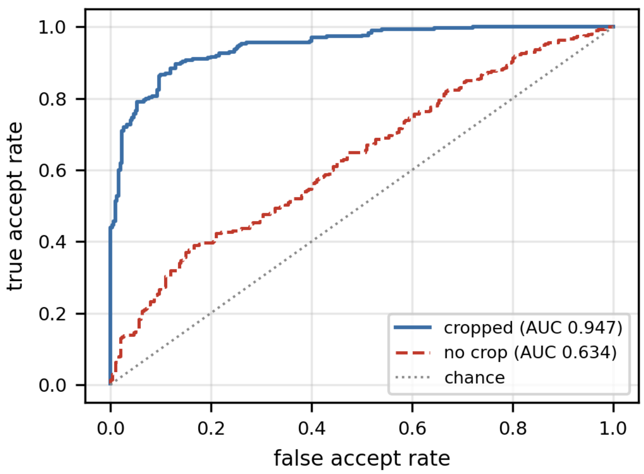

Fig. 2. ROC on the held-out test split, with and without face cropping.

Two figures, one task. Table III reports 54.7% at $N = 150$ while the headline figure is 58.7%. These measure different things and both are correct. The headline is the accuracy of the 150-identity gallery the released harness enrols, which is the number the code prints. The table entry is the mean over 200 randomly drawn sub-galleries of the same size, which is the right estimator for a *curve* because it removes the luck of any single draw. The gap of 4.0 points is close to the ± 3.1 standard deviation across those draws, which is itself the point: at this gallery size a single draw routinely lands several points either side of the mean.

Eq. (2) tracks measurement to within 2.8 points up to $N = 150$; Eq. (3) is wrong by a factor of 55 there and by three orders of magnitude at $N = 300$.

This is a consistency check, not a held-out prediction. The p_i are computed from the same 300×300 distance matrix whose sub-galleries the equations then describe, so Eq. (2) has no free parameters and no unseen data; it is a re-description of that matrix under an independence assumption. Its value is therefore not the closeness of the fit but the *structure of the residual*, which tests the assumption directly. Two observations follow. The binomial standard error at a single N is about 4 points at $N = 150$, so no individual row carries weight. And the residuals are systematic rather than random: Eq. (2) under-predicts at $N = 2$ and $N = 5$, is exact at $N = 10$, and over-predicts from $N = 25$ onward with the error growing monotonically from $+0.6$ to $+5.2$ points. That is the signature of positive dependence among distractors, or hubness [8], which is precisely the assumption Eq. (2) *violates*. The model is a good first-order account, not an exact one, and it errs in the direction its assumption predicts.

Eq. (3) fails because $\bar{p} = 3.1\%$ describes no actual probe. The distribution is bimodal: 140 of 300 probes have p_i exactly zero, 111 fall in $(0, 0.05)$ and 49 exceed 0.05. The zero-error probes alone contribute 46.7 points of the 54.7 resampled mean at $N = 150$; almost all of the remaining 8.0 comes from the thin band with small non-zero p_i, since $(1 - p_i)^{149}$ has already decayed to e^{-1} at $p_i = 1/149$. Averaging $\bar{p}$ destroys exactly this structure.

D. Open-set identification is much harder

Adding 300 never-enrolled probes changes the picture sharply. On a fixed 150-identity gallery, whose closed-set rank-1 is 58.7%, the DIR is 14.7% at 1% FAR and 36.0% at 10% FAR. Reaching the closed-set figure requires $\tau \approx 0.9$, at which 94.7% of strangers are wrongly accepted. The closed-set ceiling quoted here refers to that single fixed gallery, not to the 200-sub-gallery average of Table III.

Open-set requires two conditions at once: the correct identity must win the 149-way comparison *and* the winning distance must clear a threshold tight enough to exclude strangers. Those requirements pull τ in opposite directions, which is why the operationally relevant number is roughly a quarter of the closed-set figure and a sixth of the verification figure.

E. Multi-shot enrolment

The model predicts that reducing genuine distance, rather than altering the matching rule, is what should help. We

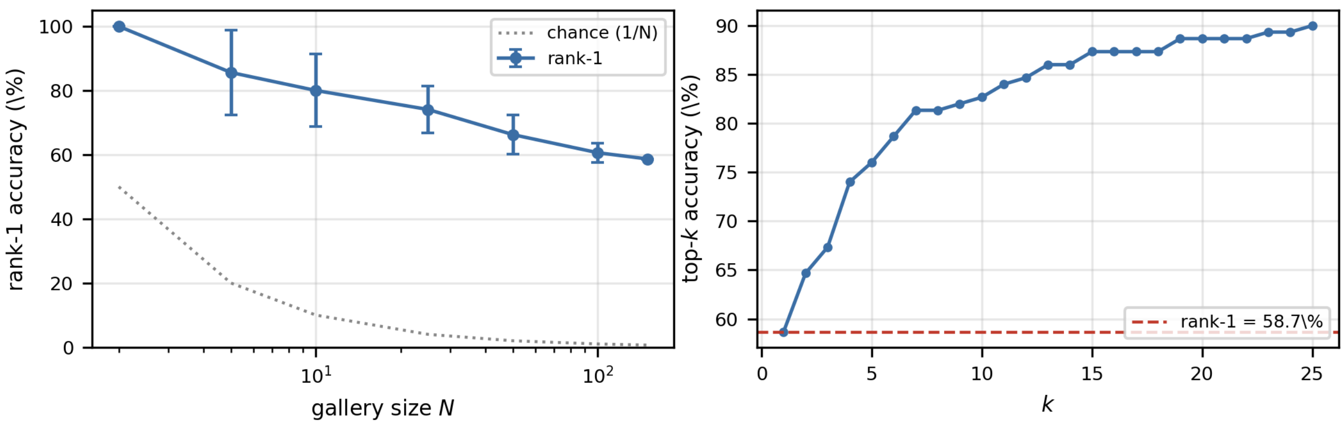

Fig. 3. Left: measured closed-set rank-1 against gallery size, with both models overlaid. The logarithmic ordinate is needed to contain the collapse of Eq. (3), which compresses the measured points and Eq. (2) into the top of the axis; Table III is the better reference for the size of that residual. Right: open-set operating curve for the fixed 150-identity gallery.

TABLE IV
MULTI-SHOT ENROLMENT, 150 IDENTITIES WITH AT LEAST FOUR PHOTOGRAPHS. THE IMPOSTOR BAR IS REPORTED ALONGSIDE TO SHOW THE GAIN IS NOT A GENERAL CONTRACTION OF DISTANCES.

k	Rank-1	Genuine d	$\mathbb{E}[\min d_{\text{imp}}]$	Margin
1	52.0%	0.743	0.735	-0.008
2	62.7%	0.666	0.715	$+0.049$
3	**69.3%**	**0.627**	0.708	$\mathbf{+0.081}$

enrolled each identity with the L^2-renormalised mean of k embeddings, holding gallery size and probe fixed.

Table IV confirms the prediction and isolates the mechanism. Averaging does shrink distances for impostor probes too, so the bar moves, from 0.735 to 0.708. But the genuine distance falls four times as far, from 0.743 to 0.627, and the margin flips from -0.008 to $+0.081$. The gain is therefore probes crossing the bar, not a uniform contraction. It requires no change to the network, threshold or matching rule. The single-shot baseline is 52.0% rather than the pooled figure because this experiment draws from the 610 identities with at least four photographs.

VI. DISCUSSION

A. What drives failure

Successful probes match their own second photograph at mean distance 0.58, below the bar; failing probes match at 0.95, above it. Rank correlates with genuine distance at Spearman $\rho = 0.88$. This is partly definitional, since rank counts distractors inside the genuine radius, so we decompose it. Local impostor density, the mean distance to a probe's five nearest wrong identities, is essentially uncorrelated with rank ($\rho = -0.03$, $p = 0.56$), and the partial correlation of genuine distance with rank controlling for density is $+0.93$. The impostor cloud is therefore close to homogeneous across probes, and rank is governed almost entirely by how far a person's own two photographs land apart. Failure is a property of the probe, not of an unlucky neighbourhood.

Consistent with this, the correct identity usually remains near the top. Using 1-indexed ranks, the median rank of the true identity is 2, with top-5 accuracy 71.0% and top-25 accuracy 88.7% against rank-1 of 46.7% on the full 300-identity pool.

B. A negative result on automatic detection

Because a fixed crop cannot adapt to off-centre faces, we replaced it with OpenCV's YuNet CNN detector, cropping to the detected box. The detector fired on 100% of images, yet accuracy fell. Under our earlier protocol, a single 600-pair sample and one fixed 150-identity gallery, the fixed crop gave 87.3% verification and 58.7% rank-1 against the detector's 85.2% and 42.0%.

This comparison is weaker than it appears and we flag it as such. The fixed crop received a nine-candidate search on a disjoint tuning split, whereas the detector box received only a coarse search, on the same pairs it was scored on, over box scale and vertical shift. So this is a carefully tuned crop against a casually tuned detector, which is exactly the asymmetry this paper criticises elsewhere. We therefore offer, rather than establish, the following explanation: the `lfw-deepfunneled` images are already aligned, so a second detector cannot remove misalignment that has largely been eliminated, and can only substitute a different box convention while adding per-image jitter. That jitter cost identification 16.7 points against verification's 2.1, an asymmetry consistent with Section V-C. This experiment is not reproducible from the submitted artifact, as noted in Section I.

One incidental observation supports the noise argument of Section V-A. The fixed crop scores 87.3% under this earlier protocol and 89.2% on the test split: the same crop, the same model, differing only in which 600 pairs were drawn, moves 1.9 points. That is the scale of variation against which the 0.9-point spread among the shifted windows should be judged.

C. Threats to validity

The crop constant predates the tuning/test partition and was originally fixed on LFW pairs not guaranteed disjoint from the present test split, so some optimism may remain in the headline figure despite the validation of Section V-A; the two splits also share 99 images, and the size choice within the shifted family is noise-dominated. Our ten folds are random partitions of a 600-pair sample rather than LFW's predefined identity-disjoint folds, so the same person can appear in both the threshold-fitting and scored folds; this is a mild optimism leak that a strict protocol would remove. We evaluate 600 of 3,000 available pairs per half for compute reasons, which is the direct cause of the wide intervals. Verification runs on the LFW pair list while identification runs only on multi-photograph identities, so differences between tasks are attributable to task *and* probe population, not task alone. LFW consists largely of frontal, well-lit press photographs, so these figures are an upper bound on unconstrained performance. Finally, we performed no demographic subgroup analysis, and aggregate accuracy is known to conceal substantial disparities across groups [11].

VII. Conclusion and Future Work

On one frozen embedding we measured 89.2% verification, 58.7% closed-set rank-1 on a 150-identity gallery, and 14.7% open-set DIR at 1% FAR. The spread is not a defect of the model but a consequence of what each task requires. Closed-set identification must beat the nearest of $N-1$ competitors, a bar falling from 1.28 to 0.70 as the gallery grows from 2 to 300, and a per-probe order-statistics model reproduces the measured curve to within 2.8 points while erring in the direction its independence assumption predicts. Open-set additionally demands that the winning distance clear a threshold tight enough to reject strangers, costing a further 44 points.

Preprocessing, not architecture, was the dominant implementation variable, and within it the vertical placement of the crop mattered as much as its size, while the size itself proved noise-dominated.

The failure analysis indicates where to go next. Because failure concentrates in probes with large genuine distances, and local impostor density is essentially irrelevant, anything reducing that distance pays disproportionately. Multi-shot enrolment demonstrates this, worth 17.3 points with the impostor bar barely moving and without any architectural change, and landmark-based alignment attacks the same quantity by normalising pose. A backbone trained with angular-margin losses [10] would raise the ceiling, but should be expected to shift the curve rather than change its shape.

Practically, identification results should always be reported with their gallery size and with whether they are open or closed set, and verification and identification figures should never be quoted interchangeably. A single unqualified accuracy number is not a meaningful description of a face recognition system.

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